Hochschule der Bayerischen Wirtschaft

Registration final module Master´s thesis

Title of the Master´s thesis: **A Supervised Dimensionality-Reduction and Temporal Deep-Learning Framework for Attack Detection in EV Charging Infrastructure**

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SYNOPSIS

A Supervised Dimensionality-Reduction and Temporal Deep-Learning Framework for Attack Detection in EV Charging Infrastructure

1. Introduction

The rapid deployment of Electric Vehicle (EV) charging infrastructure has transformed EV charging stations into important components of modern cyber-physical systems. EV charging stations, commonly implemented using Electric Vehicle Supply Equipment (EVSE), communicate with charging management systems and electric vehicles through communication protocols and interfaces such as the Open Charge Point Protocol (OCPP) and ISO 15118.

The increasing connectivity and computational capabilities of EV charging infrastructure introduce new cybersecurity risks. An attacker may exploit communication interfaces, network services, operating-system resources, or charging-station software to perform reconnaissance, denial-of-service (DoS), cryptojacking, backdoor and other malicious activities.

Traditional intrusion detection systems generally depend on predefined signatures or manually engineered rules. Such approaches may have difficulty identifying complex or evolving attack behaviour. Consequently, machine-learning and deep-learning-based intrusion detection systems have received increasing attention for EV charging security.

The **CICEVSE2024 dataset**, developed by the Canadian Institute for Cybersecurity at the University of New Brunswick, provides a comprehensive benchmark for investigating cybersecurity attacks against EV charging infrastructure. The dataset contains benign and malicious scenarios collected from an EVSE testbed and includes three major categories of telemetry: **network traffic, HPC/kernel events, and power-consumption data**. The attack scenarios include reconnaissance and DoS attacks at the network level and backdoor and cryptojacking attacks at the host level.

The availability of high-dimensional host-level measurements creates an opportunity to investigate supervised dimensionality-reduction techniques before applying temporal deep-learning models. This research therefore proposes a systematic comparison of LSTM-based and hybrid temporal deep-learning architectures with and without dimensionality reduction.

The final objective is to select the best-performing and computationally suitable model and deploy it on an **NVIDIA Jetson Nano**, representing a resource-constrained edge/end-point device located close to an EV charging station. The deployed model will perform attack detection locally and provide a basis for real-time or near-real-time EVSE security monitoring.

2. Problem Statement

EV charging infrastructure generates large volumes of heterogeneous cybersecurity telemetry. In particular, the CICEVSE2024 host-level dataset contains a large number of Hardware Performance Counter (HPC) and kernel-event features. These high-dimensional feature spaces may contain redundant, correlated and less informative variables, increasing the computational requirements of deep-learning models.

At the same time, cyberattacks against EVSE can exhibit temporal characteristics. For example, reconnaissance and DoS attacks may generate behavioural patterns that evolve over a sequence of observations rather than being identifiable from a single observation.

Therefore, an effective EVSE intrusion detection system should address two major challenges:

1. **High-dimensional feature representation**, and
2. **Temporal attack-pattern detection**.

This research investigates whether supervised and unsupervised dimensionality-reduction techniques can improve the performance and computational efficiency of temporal deep-learning models for EVSE attack detection.

The study will compare five deep-learning pipelines:

LSTM without dimensionality reduction

PCA-LSTM

LDA-LSTM

LDA-BiLSTM

CNN-BiLSTM with Attention

The models will be evaluated using common classification and computational-performance metrics and the best-performing model will subsequently be optimized and deployed on a Jetson Nano edge device.

3. Literature Survey

Recent studies have demonstrated the applicability of deep learning to cybersecurity attack detection in EV charging infrastructure using the CICEVSE2024 dataset. Transfer-learning approaches have investigated LSTM, RNN, GRU and DNN architectures for cyber-physical attack detection, while more recent studies have explored Transformer-based transfer learning and reported high classification performance. Graph-based approaches have further investigated relationships among HPC and kernel-event features through graph neural networks and graph contrastive learning. Multi-task learning has also been proposed to jointly address binary, attack-family and fine-grained attack classification. These studies demonstrate the effectiveness of advanced deep-learning techniques but leave scope for a systematic investigation of supervised dimensionality reduction, particularly LDA, in combination with temporal deep-learning architectures. Therefore, this thesis investigates LSTM, PCA-LSTM, LDA-LSTM, LDA-BiLSTM and CNN-BiLSTM-Attention using consistent evaluation criteria, followed by optimization and deployment of the selected model on an NVIDIA Jetson Nano.

4. Research gaps

Gap 1 : No systematic dimensionality-reduction comparison

The existing research work performs correlation analysis and selects the top 10 features. It does not systematically investigate PCA or LDA as dimensionality-reduction techniques before temporal deep learning. This allows you to answer whether unsupervised PCA or supervised LDA provides a more useful representation for attack detection.

Gap 2 : No systematic comparison of temporal DL architectures

The existing research work does not provide a controlled comparison between:

A. LSTM
B. PCA-LSTM
C. LDA-LSTM
D. LDA-BiLSTM
E. CNN-BiLSTM + Attention

Gap 3 : Limited attack scope

The existing implementation focuses specifically on DDoS detection and can be

extended to Reconnaissance, DoS, Cryptojacking and Backdoor.

Gap 4: No edge-device deployment

The attached report develops and evaluates the model but does not demonstrate deployment of the trained IDS on an edge device. This converts the research from a dataset-only experiment into a prototype edge-based EVSE attack-detection system.

5. Aim of the Research

The primary aim of this research is:

To develop and experimentally evaluate a dimensionality-reduction and temporal deep-learning framework for detecting cyberattacks in EV charging infrastructure using the CICEVSE2024 dataset, followed by deployment of the best-performing model on an NVIDIA Jetson Nano edge device.

6. Research Objectives

The objectives of the proposed research are:

Objective 1

To preprocess and prepare the CICEVSE2024 dataset for temporal deep-learning-based EVSE attack detection.

Objective 2

To investigate PCA and LDA as dimensionality-reduction techniques for EVSE cybersecurity data.

Objective 3

To develop and compare the following five pipelines:

A. LSTM

B. PCA-LSTM

C. LDA-LSTM

D. LDA-BiLSTM

E. CNN-BiLSTM + Attention

Objective 4

To evaluate the models using Accuracy, Precision, Recall, F1-score, ROC-AUC and Training Time, and select a model considering both detection performance and computational requirements.

Objective 5

To optimize and deploy the selected model on an NVIDIA Jetson Nano and evaluate its feasibility for edge-based EVSE attack detection.

7. Research Questions

The research will address the following questions.

RQ1 : Dimensionality reduction

How does dimensionality reduction using PCA and LDA affect the attack-detection performance and computational requirements of temporal deep-learning models for EV charging infrastructure?

RQ2 : Temporal deep-learning architectures

How do LSTM, BiLSTM, and CNN-BiLSTM with Attention compare in detecting benign and attack patterns in the CICEVSE2024 dataset?

RQ3 : Performance and efficiency

Which of the investigated pipelines provides the most effective trade-off between detection performance and computational cost based on Accuracy, Precision, Recall, F1-score, ROC-AUC, and Training Time?

RQ4 : Edge deployment

To what extent can the selected attack-detection model be optimized and deployed on an NVIDIA Jetson Nano for edge-based EV charging infrastructure attack detection?

8. Dataset

The proposed research will use the CICEVSE2024 dataset, published by the Canadian Institute for Cybersecurity at the University of New Brunswick.

The dataset was generated using an EV charging testbed containing EVSE components, Raspberry Pi-based devices, communication infrastructure and power-monitoring equipment. The dataset includes both benign and attack scenarios under idle and charging conditions.

The principal data sources are:

6.1 Network Traffic

Network traffic associated with EVSE-A and EVSE-B is provided through packet captures and processed traffic data.

The testbed includes OCPP communication with the Charging Station Monitoring System (CSMS) and ISO 15118 communication between the EV communication controller and EVSE.

6.2 HPC and Kernel Events

The dataset contains host-level measurements collected from EVSE-B. Approximately 900 HPC/kernel events have been used in research involving the dataset, providing information about CPU, memory, system-call and other low-level system behaviour.

6.3 Power Consumption

Power-consumption measurements from EVSE-B provide a cyber-physical view of the charging station's behaviour.

6.4 Attack Categories

The dataset contains attack scenarios including:

- Reconnaissance
- Denial of Service (DoS)
- Cryptojacking
- Backdoor

Specific attack examples include port scanning, service detection, OS fingerprinting, vulnerability scanning, SYN flooding, TCP flooding, ICMP flooding and related attacks.

The dataset also provides labels such as:

- State
- Scenario
- Attack
- Label
- Interface

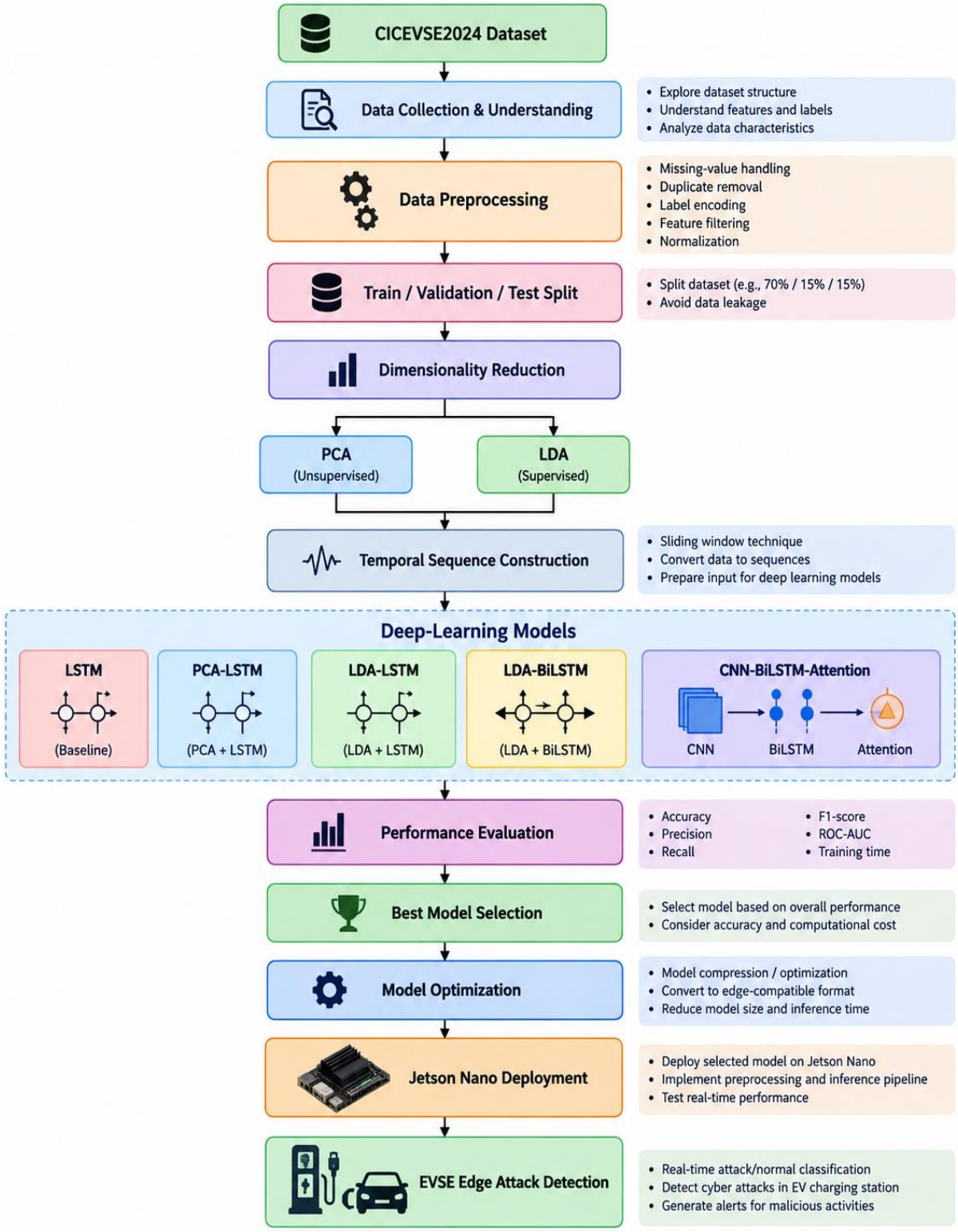
where the primary binary label distinguishes Attack and Benign instances.

9. Proposed Methodology

The proposed methodology consists of six major stages:

1. Dataset Collection & Preprocessing
2. Feature Engineering & Dimensionality Reduction

3. Temporal Sequence Construction
4. Deep Learning Model Development
5. Performance Evaluation & Best-Model Selection
6. Jetson Nano Edge Deployment & Validation


CICEVSE2024 Dataset
Data Collection & Understanding
• Explore dataset structure
• Understand features and labels
• Analyze data characteristics
Data Preprocessing
• Missing-value handling
• Duplicate removal
• Label encoding
• Feature filtering
• Normalization
Train / Validation / Test Split
• Split dataset (e.g., 70% / 15% / 15%)
• Avoid data leakage
Dimensionality Reduction
PCA
(Unsupervised)
LDA
(Supervised)
Temporal Sequence Construction
• Sliding window technique
• Convert data to sequences
• Prepare input for deep learning models
Deep-Learning Models
LSTM
(Baseline)
PCA-LSTM
(PCA + LSTM)
LDA-LSTM
(LDA + LSTM)
LDA-BiLSTM
(LDA + BiLSTM)
CNN-BiLSTM-Attention
CNN
BiLSTM
Attention
Performance Evaluation
• Accuracy
• Precision
• Recall
• F1-score
• ROC-AUC
• Training time
Best Model Selection
• Select model based on overall performance
• Consider accuracy and computational cost
Model Optimization
• Model compression / optimization
• Convert to edge-compatible format
• Reduce model size and inference time
Jetson Nano Deployment
• Deploy selected model on Jetson Nano
• Implement preprocessing and inference pipeline
• Test real-time performance
EVSE Edge Attack Detection
• Real-time attack/normal classification
• Detect cyber attacks in EV charging station
• Generate alerts for malicious activities

10. Data Preprocessing

The raw CICEVSE2024 data will undergo preprocessing before model development. The preprocessing stages will include:

8.1 Data Cleaning

- Identification of missing values
- Handling of infinite values
- Removal of irrelevant fields
- Removal of duplicate observations where appropriate
- Identification of invalid or corrupted records

8.2 Feature Processing

Categorical variables will be appropriately encoded, while numerical features will be normalized or standardized.

Features that directly reveal the target label or attack identity will not be used as predictive inputs in order to prevent target leakage.

8.3 Label Definition

The primary detection task will initially be defined as binary classification:

```
0 → Benign
1 → Attack
```

A secondary experiment may investigate multi-class attack classification where computationally and methodologically appropriate.

11. Train/Test Strategy

Special attention will be given to avoiding data leakage.

Because CICEVSE2024 contains temporally collected observations and attack scenarios captured over extended periods, a simple random row-wise split may produce overly optimistic results if highly similar observations from the same attack session occur in both training and testing sets. The dataset documentation notes that the original CSV files contain samples captured for each scenario over extended periods.

Therefore, where the data organization permits, the research will use a scenario/session-aware or chronological split.

The proposed arrangement is:

```
Training data
      │
      ├── Fit scaler
      ├── Fit PCA
      └── Fit LDA

Validation data
      │
      └── Transform using training parameters

Test data
      │
      └── Transform using training parameters
```

Importantly, PCA and LDA will be fitted only on the training data.

This prevents information from the test set from influencing the dimensionality-reduction transformation.

12. Temporal Sequence Construction

Since LSTM and BiLSTM models require sequential input, the preprocessed observations will be transformed into temporal windows.

For example:

```
X1 X2 X3 X4 X5 X6 X7 X8 X9 ...
|──────────────|
    Window 1

   X2 X3 X4 X5 X6 X7 X8 X9 X10
   |──────────────|
       Window 2
```

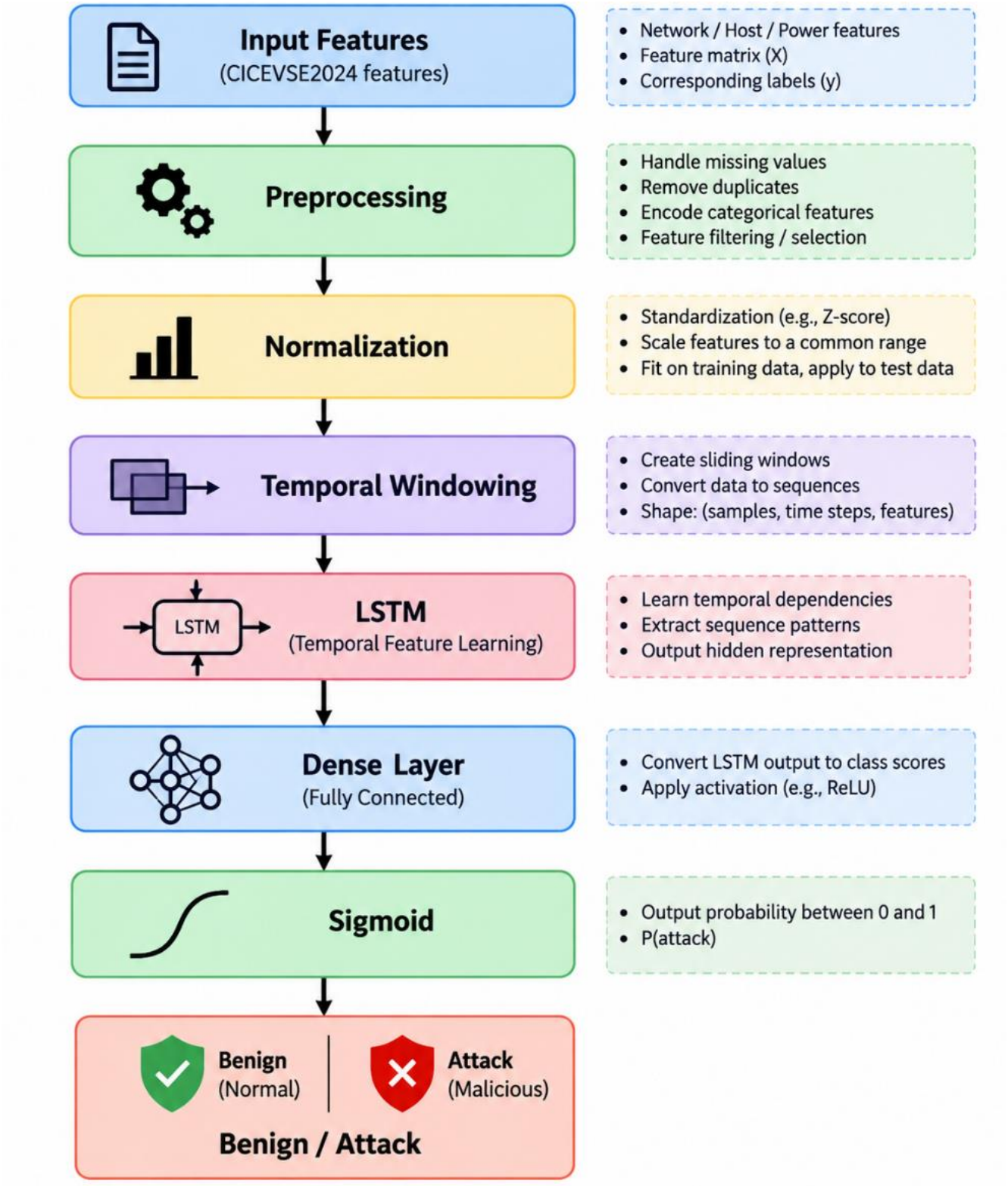
Each temporal window will contain a sequence of feature vectors.

The appropriate window size will be experimentally determined using the temporal characteristics and sampling rate of the selected CICEVSE2024 data source.

13. Proposed Experimental Pipelines

Pipeline A : LSTM without LDA

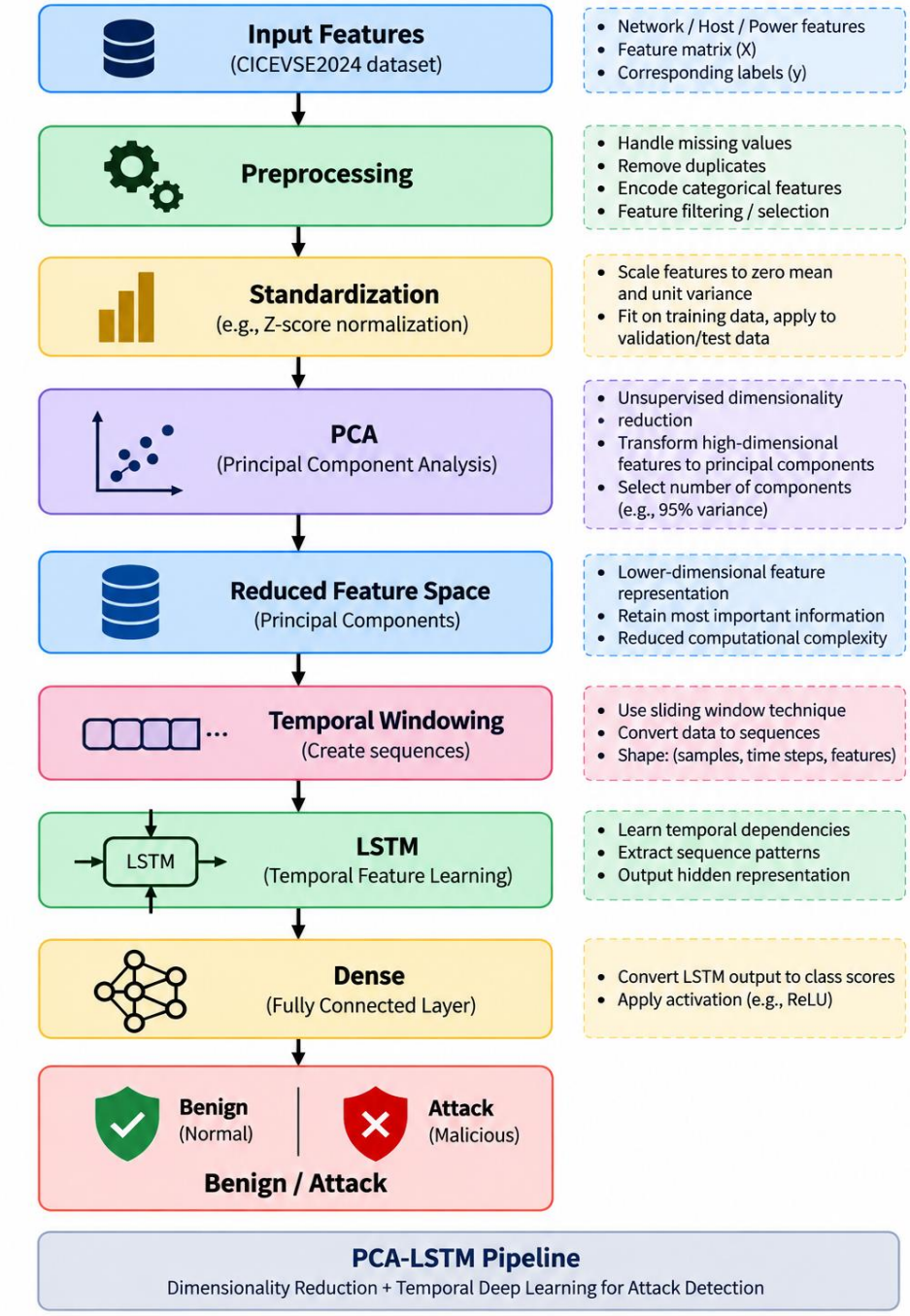
This will serve as the principal baseline.



The purpose is to establish the performance of temporal deep learning without dimensionality reduction.

Pipeline B : PCA-LSTM

The second experiment will investigate unsupervised dimensionality reduction.

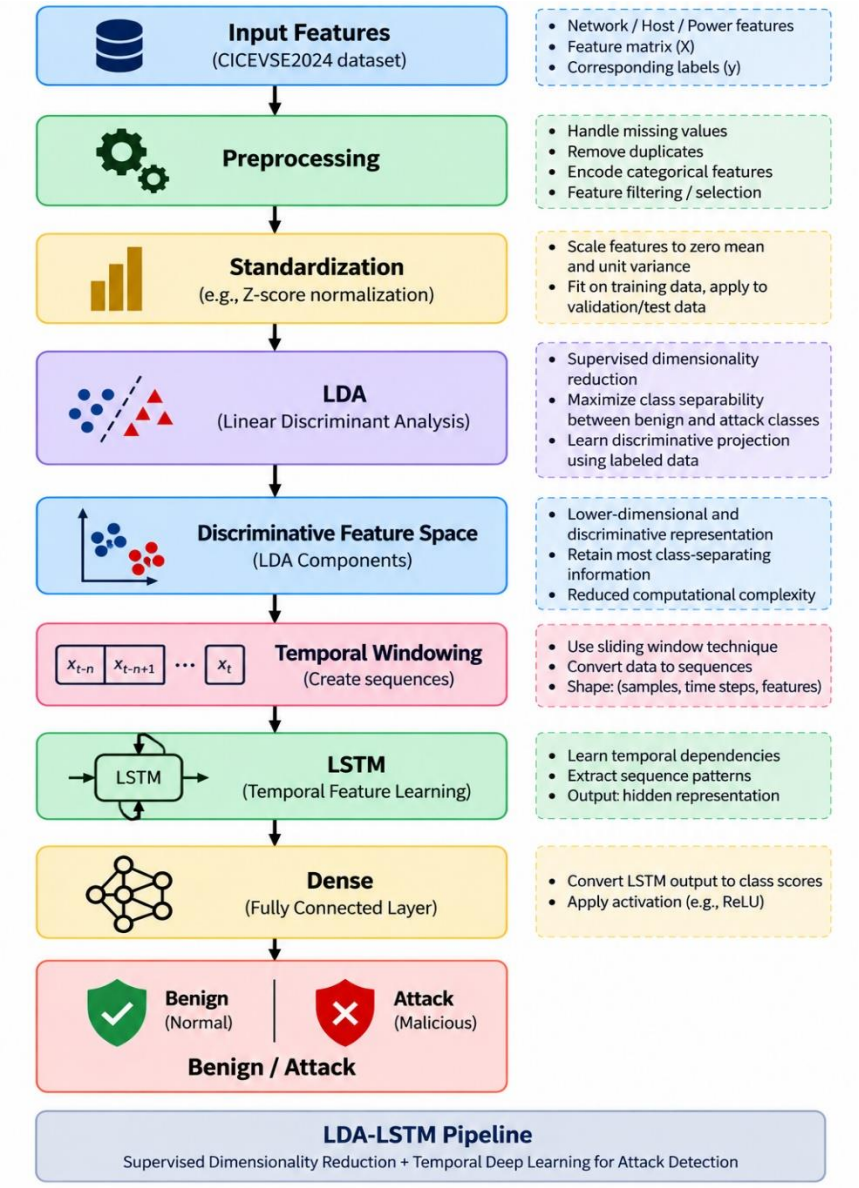


The number of principal components will be selected using explained-variance analysis and/or validation performance.

This experiment will determine whether redundant feature removal improves the LSTM model.

Pipeline C : LDA-LSTM

The third pipeline will investigate supervised dimensionality reduction.

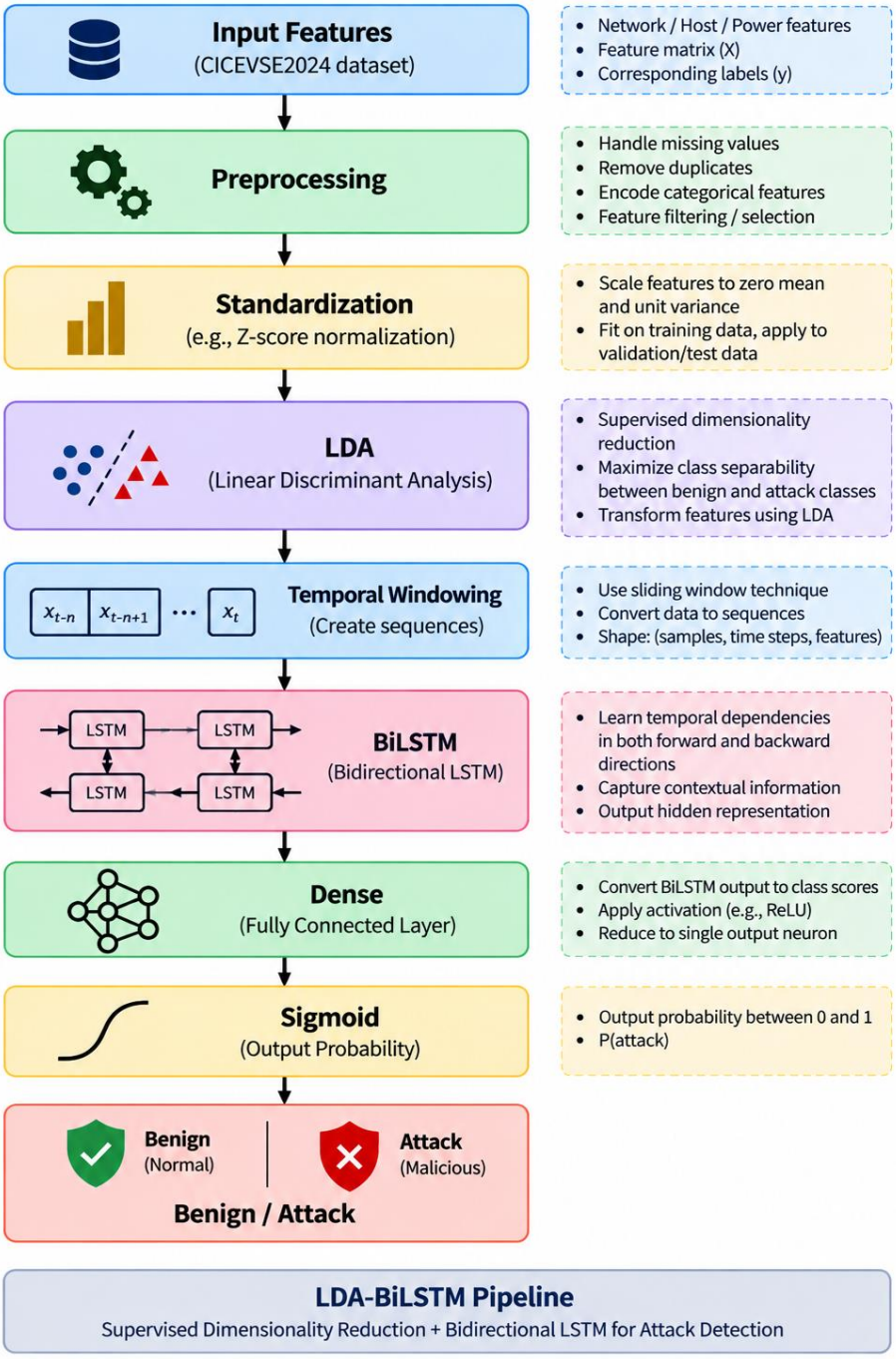


Unlike PCA, LDA uses class information to maximize class separability. This is particularly relevant to the proposed research because the objective is to determine whether a supervised projection can produce a more attack-discriminative feature representation.

A limitation will be explicitly considered: for binary classification, classical LDA provides at most C−1 = 1 discriminant component. Therefore, the thesis will experimentally verify whether this aggressive dimensionality reduction retains sufficient temporal information. If necessary, the multi-class task or alternative feature-selection strategies will be investigated as a supplementary experiment.

Pipeline D : LDA-BiLSTM

The fourth pipeline extends the LDA-LSTM architecture by replacing LSTM with Bidirectional LSTM.

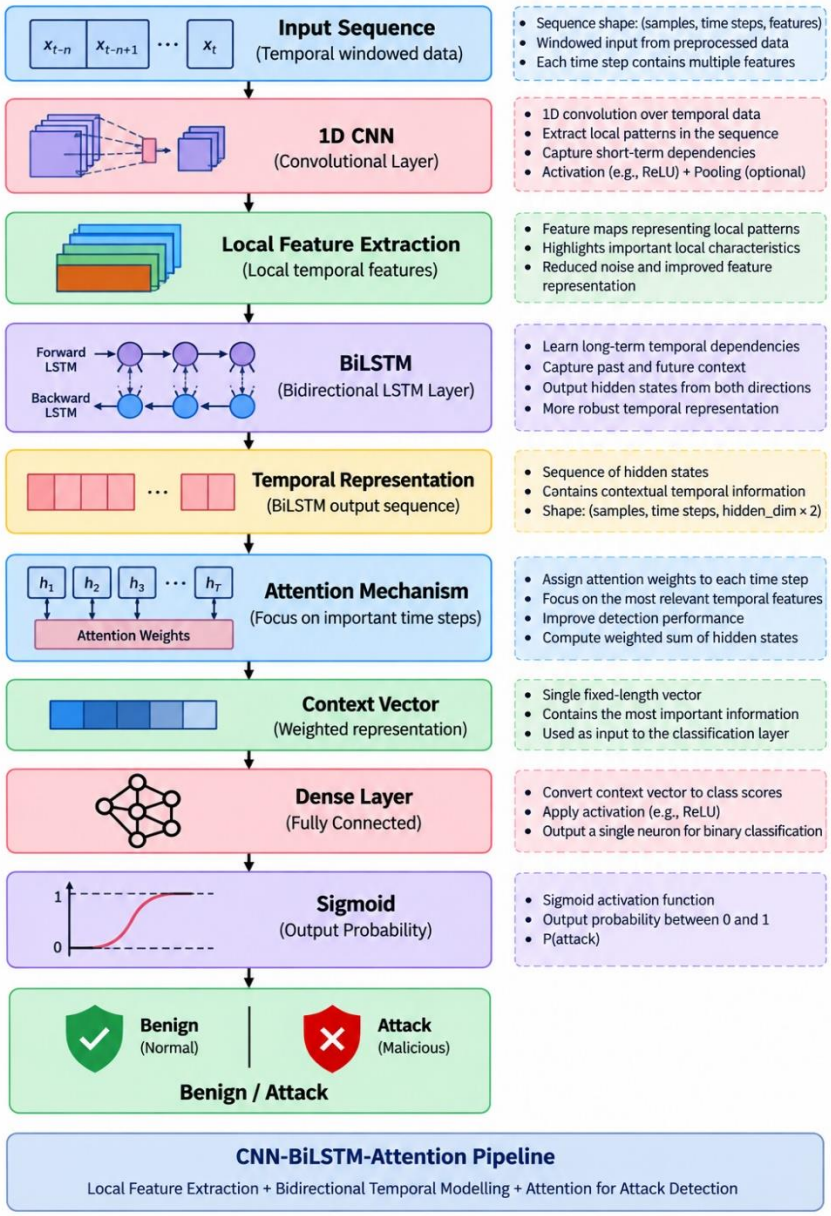


BiLSTM processes the sequence in both forward and backward directions within each available temporal window.

The purpose is to investigate whether bidirectional temporal representation provides better discrimination between normal and attack sequences.

Pipeline E : CNN-BiLSTM + Attention

The fifth and most sophisticated experimental architecture will combine local feature extraction, bidirectional temporal modelling and attention.



The CNN will identify local patterns within the feature sequences.

The BiLSTM will model temporal dependencies.

The attention mechanism will assign greater importance to informative temporal representations.

This model will be evaluated against the simpler LSTM and dimensionality-reduction-based models to determine whether the additional architectural complexity provides a measurable improvement.

14. Comparative Experimental Framework

The five models will be evaluated under consistent experimental conditions.

Experiment	Dimensionality Reduction	Deep Learning Model
A	None	LSTM
B	PCA	LSTM
C	LDA	LSTM
D	LDA	BiLSTM
E	None / selected feature representation	CNN-BiLSTM + Attention

Where possible, the same train/test protocol, preprocessing procedure, temporal-window strategy and evaluation criteria will be used across all experiments.

15. Evaluation Metrics

The proposed models will be evaluated using the following metrics.

13.1 Accuracy

Measures the proportion of correctly classified instances.

13.2 Precision

Measures how many instances predicted as attacks are actually attacks.

13.3 Recall

Measures the ability to identify actual attacks.

Recall is particularly important for cybersecurity because missed attacks represent false negatives.

13.4 F1-score

F1-score is a widely used evaluation metric for classification problems, particularly in cybersecurity and intrusion detection. It provides a balance between Precision and Recall.

13.5 ROC-AUC

The Receiver Operating Characteristic Area Under the Curve will be used to evaluate discrimination between benign and attack classes over different decision thresholds.

13.6 Training Time

Training time will be measured to assess computational efficiency.

13.7 Additional Edge Metrics

For the final Jetson Nano deployment, the following metrics will additionally be measured:

- Inference latency
- Inference throughput
- CPU utilization
- GPU utilization
- RAM usage
- Model size
- Power consumption, where measurable
- End-to-end detection latency

These metrics are important because the objective is not only to obtain high classification accuracy but also to determine whether the model is practical for deployment on a resource-constrained endpoint device.

16. Model Selection

The best model will not necessarily be selected solely according to accuracy.

A multi-criteria selection process will consider:

```
Detection Performance
          +
      Recall/F1
          +
       ROC-AUC
          +
   Training Cost
          +
  Inference Latency
          +
  Memory Requirement
          +
Jetson Nano Compatibility
```

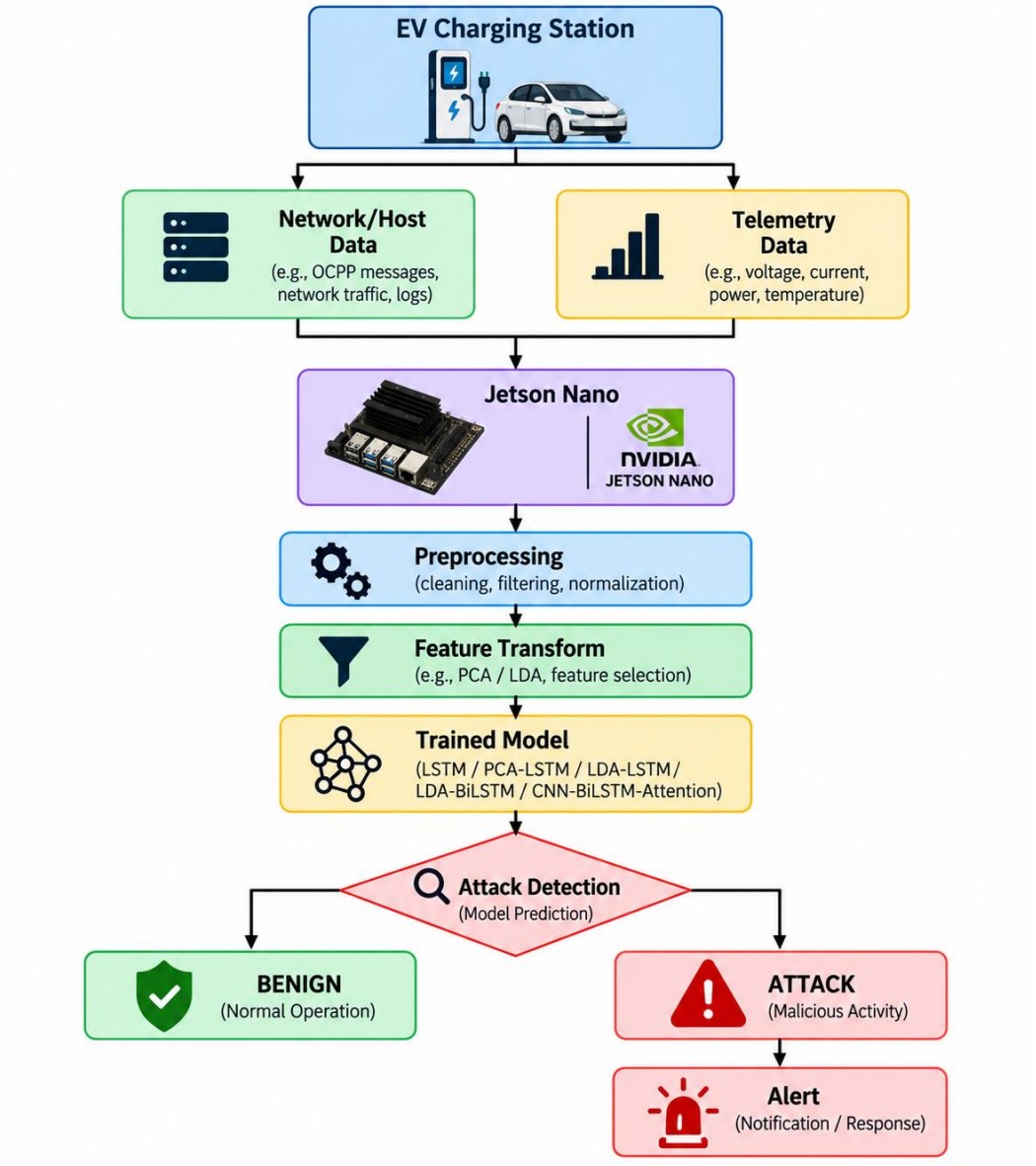
For example, if two models achieve similar F1-scores but one has substantially lower inference latency and memory consumption, the smaller model may be preferable for edge deployment.

17. Jetson Nano Deployment

After evaluating the five models, the best-performing model will be selected for deployment.

The NVIDIA Jetson Nano will act as an endpoint/edge computing device representing a security monitoring component associated with an EV charging station.

The proposed deployment architecture is:



The same preprocessing transformations used during model training must be preserved for deployment.

The deployment experiment will demonstrate whether the selected model can perform inference within the resource constraints of an edge device.

18. Proposed Software and Hardware Environment

16.1 Software

The implementation may use:

- Python
- Pandas
- NumPy
- Scikit-learn
- TensorFlow/Keras or PyTorch
- Matplotlib
- Seaborn, where appropriate for exploratory analysis
- ONNX/TensorRT where required for optimized Jetson deployment

16.2 Development Environment

Model training will preferably be performed on a workstation/GPU environment because CICEVSE2024 contains large volumes of data.

16.3 Edge Environment

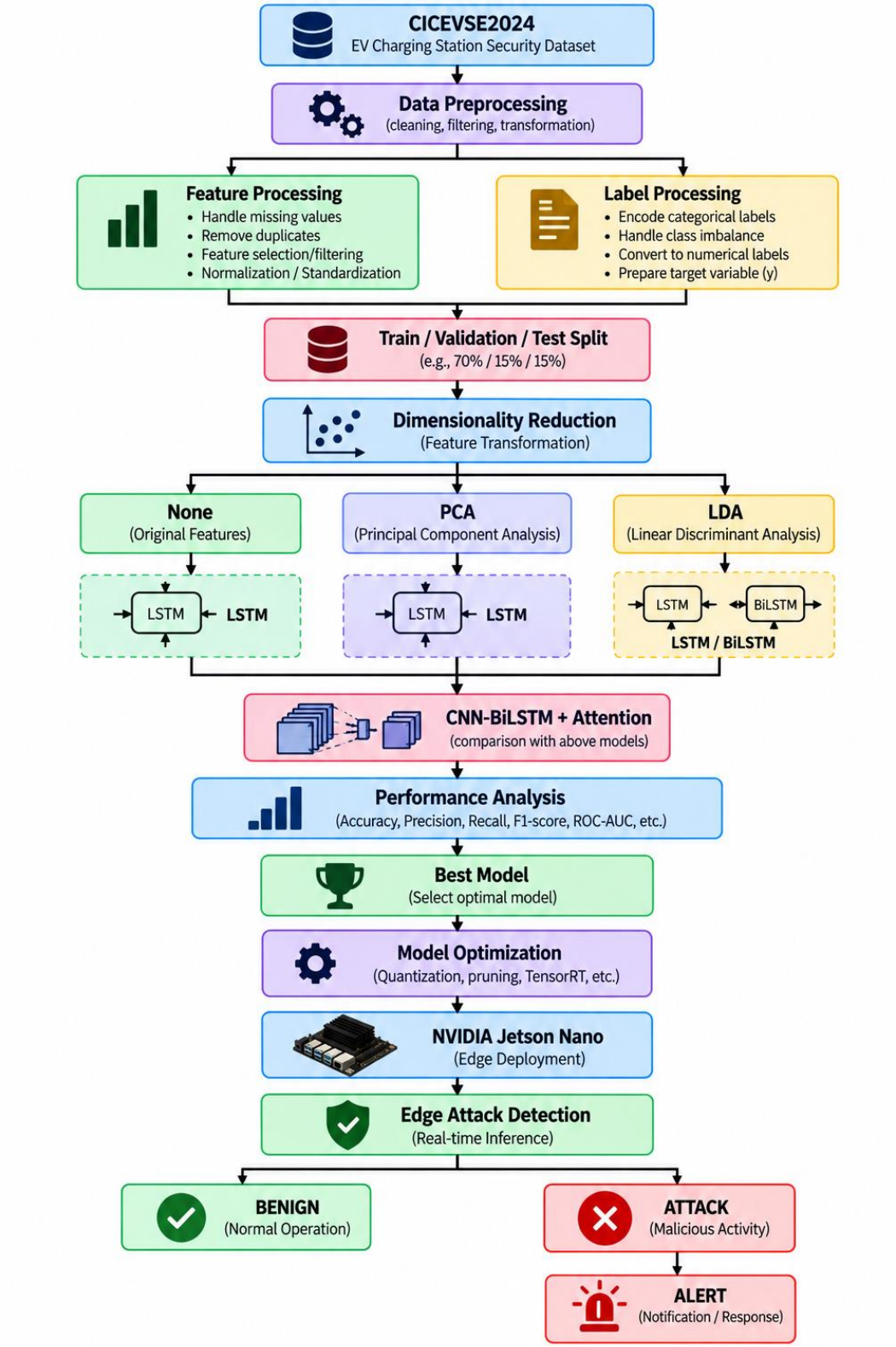
The final model will be deployed on:

NVIDIA Jetson Nano

The deployment environment will use an appropriate JetPack/CUDA/TensorRT-compatible software stack.

19. Proposed System Architecture

The complete proposed system is:



20. Expected Contributions

The expected contributions of this research are:

Contribution 1 : Dimensionality-reduction investigation

A systematic investigation of PCA and supervised LDA for reducing the dimensionality of CICEVSE2024 EVSE cybersecurity data and analyzing their impact on temporal attack detection.

Contribution 2 : Comparative temporal deep-learning framework

A controlled experimental comparison of LSTM, PCA-LSTM, LDA-LSTM, LDA-BiLSTM and CNN-BiLSTM-Attention architectures for EVSE attack detection.

Contribution 3 : Comprehensive performance and efficiency evaluation

A comparative evaluation of the investigated pipelines using Accuracy, Precision, Recall, F1-score, ROC-AUC and Training Time, complemented by computational and inference measurements for the selected model.

Contribution 4 : Edge-based EVSE attack detection prototype

An optimized implementation of the selected attack-detection model on an NVIDIA Jetson Nano to demonstrate the feasibility of edge-based EVSE security monitoring.

21. Expected Outcomes

The expected outcomes are:

1. Identification of the most suitable temporal deep-learning architecture for CICEVSE2024-based attack detection.
2. Quantification of the effect of PCA and LDA on feature dimensionality.
3. Determination of whether supervised LDA provides an advantage over PCA.
4. Identification of the performance difference between LSTM and BiLSTM.
5. Determination of whether CNN-BiLSTM-Attention provides a meaningful improvement over simpler architectures.
6. Identification of the trade-off between detection accuracy and computational complexity.
7. Development of a trained EVSE attack-detection model suitable for edge deployment.
8. Demonstration of attack detection on an NVIDIA Jetson Nano endpoint.
9. Measurement of inference latency and resource utilization on the edge device.
10. Development of a foundation for real-time or near-real-time EV charging cybersecurity monitoring.

22. Scope of the Research

The primary scope of the research is attack-versus-benign detection using the CICEVSE2024 dataset.

The initial experimental problem will therefore be:

```
Input:
EVSE telemetry

Output:
Benign
or
Attack
```

The dataset's attack and scenario labels also provide an opportunity for supplementary multi-class experiments. CICEVSE2024 explicitly supports binary and multi-class classification tasks.

However, the principal contribution will remain the comparative investigation of dimensionality reduction and temporal deep-learning architectures followed by edge deployment.

The research will not claim that the Jetson Nano itself represents the complete security architecture of a commercial EV charging station. Instead, it will serve as a prototype endpoint/edge platform for demonstrating the feasibility of local inference.

23. Limitations

The following limitations will be acknowledged:

1. CICEVSE2024 represents a specific experimental EVSE environment and may not cover every real-world EV charging configuration.
2. Dataset-based performance may not directly translate to operational production environments.
3. LDA is supervised and therefore depends on training labels.
4. Classical LDA has a maximum dimensionality of (C-1); therefore, binary LDA produces only one discriminant component.
5. Deep-learning models may require significant computational resources during training.
6. Jetson Nano has substantially fewer resources than a training workstation.
7. Edge deployment may require model optimization, quantization or TensorRT conversion.
8. Detection performance for previously unseen attacks will require a separate experimental design and should not automatically be interpreted as guaranteed zero-day detection.

24. Proposed Work Plan

Phase	Activity
Phase 1	Literature survey on EVSE cybersecurity and IDS
Phase 2	Detailed study of CICEVSE2024
Phase 3	Dataset acquisition and preprocessing
Phase 4	Exploratory data analysis
Phase 5	Feature selection and dimensionality analysis
Phase 6	Development of baseline LSTM
Phase 7	Development of PCA-LSTM
Phase 8	Development of LDA-LSTM
Phase 9	Development of LDA-BiLSTM
Phase 10	Development of CNN-BiLSTM-Attention
Phase 11	Comparative performance evaluation
Phase 12	Hyperparameter tuning of selected model
Phase 13	Model optimization for edge deployment
Phase 14	Jetson Nano deployment
Phase 15	Edge performance evaluation
Phase 16	Results analysis and thesis documentation

25. Conclusion

This research proposes a supervised dimensionality-reduction and temporal deep-learning framework for cybersecurity attack detection in EV charging infrastructure using the CICEVSE2024 dataset.

Five architectures will be systematically investigated: **LSTM without dimensionality reduction, PCA-LSTM, LDA-LSTM, LDA-BiLSTM, and CNN-BiLSTM with Attention**. Their effectiveness will be evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC and training time, with additional inference and resource measurements for edge deployment.

The study is motivated by the need to develop accurate yet computationally practical EVSE intrusion detection mechanisms. Existing CICEVSE2024 research has already demonstrated strong results using transfer learning, graph-based learning and other approaches, which makes a rigorous comparative and deployment-oriented study particularly relevant.

The final stage of the research will deploy the best-performing model on an **NVIDIA Jetson Nano**, demonstrating the feasibility of performing EV charging station attack detection at the edge.

The anticipated outcome is a practical and experimentally validated deep-learning-based EVSE intrusion detection prototype that balances **detection effectiveness, dimensionality, computational complexity and edge-deployment feasibility**.

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Appendix:

Timetable for the Implementation of the Scientific Work

Phase	Date / Time Period	Short Description
Milestone 1 – Thesis Registration	**15 Sep 2026 – 18 Sep 2026**	Registration of the thesis at the examination office and discussion of the research topic, objectives, methodology and work plan with the supervisor.
Planning	**19 Sep 2026 – 25 Sep 2026**	Finalization of the research problem, objectives, research questions, methodology, CICEVSE2024 dataset, experimental pipelines and preliminary thesis structure.
Research / Literature Review	**26 Sep 2026 – 09 Oct 2026**	Literature research on EV charging infrastructure cybersecurity, EVSE/OCPP security, intrusion detection, dimensionality reduction, LSTM, BiLSTM, CNN, attention mechanisms and deep-learning-based attack detection. Relevant CICEVSE2024 publications will be reviewed and structured.
Milestone 2 – Research Completed	**10 Oct 2026**	Completion of the initial literature review and research methodology. Discussion of the research gap, objectives and experimental design with the supervisor.
Dataset Preparation & Preprocessing	**10 Oct 2026 – 23 Oct 2026**	Study and preparation of CICEVSE2024. Data cleaning, missing/infinite-value handling, duplicate checking, feature analysis, label encoding, normalization/standardization and exploratory data analysis.
Milestone 3 – Dataset Ready	**24 Oct 2026**	Completion of the dataset preparation and preprocessing pipeline. Validation of the data-splitting strategy and feature preparation with the supervisor.
Pipeline A – LSTM	**24 Oct 2026 – 31 Oct 2026**	Development and training of the baseline LSTM model without dimensionality reduction. Establish baseline attack-detection performance.
Pipeline B – PCA-LSTM	**01 Nov 2026 – 07 Nov 2026**	Application of PCA for dimensionality reduction followed by LSTM. Evaluation of the effect of PCA on dimensionality, performance and training time.
Pipeline C – LDA-LSTM	**08 Nov 2026 – 14 Nov 2026**	Application of supervised LDA dimensionality reduction followed by

Phase	Date / Time Period	Short Description
		LSTM. Analysis of attack-discriminative feature representation and detection performance.
Pipeline D – LDA-BiLSTM	**15 Nov 2026 – 21 Nov 2026**	Development of the LDA-BiLSTM architecture and comparison with the LDA-LSTM model.
Pipeline E – CNN-BiLSTM + Attention	**22 Nov 2026 – 05 Dec 2026**	Development of the CNN-BiLSTM-Attention architecture combining local feature extraction, bidirectional temporal modelling and attention.
Milestone 4 – Model Development Completed	**06 Dec 2026**	All five proposed deep-learning pipelines implemented and initial results obtained. Preliminary discussion with the supervisor.
Model Evaluation & Comparison	**07 Dec 2026 – 18 Dec 2026**	Comparative evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC and Training Time. Generation of confusion matrices, ROC curves and comparative tables.
Hyperparameter Optimization	**19 Dec 2026 – 27 Dec 2026**	Optimization of the best-performing model using appropriate hyperparameter tuning. Verification of model robustness and repeatability.
Best Model Selection	**28 Dec 2026 – 31 Dec 2026**	Final selection of the best model based on detection performance and computational requirements.
Milestone 5 – Experimental Work Completed	**31 Dec 2026**	Completion of all five experiments, performance comparison and selection of the model for edge deployment. Results discussed with the supervisor.
Jetson Nano Preparation	**01 Jan 2027 – 04 Jan 2027**	Preparation of the NVIDIA Jetson Nano environment and required Python/deep-learning/optimization libraries.
Model Optimization & Conversion	**05 Jan 2027 – 10 Jan 2027**	Optimization of the selected model for edge deployment. Model conversion and compatibility testing using an appropriate Jetson-compatible inference framework.
Jetson Nano Deployment	**11 Jan 2027 – 17 Jan 2027**	Deployment of the trained model on the Jetson Nano. Implementation of preprocessing, temporal-window generation and local attack prediction.
Edge Validation & Testing	**18 Jan 2027 – 24 Jan 2027**	Testing of the deployed model and measurement of inference latency, memory consumption, CPU/GPU

Phase	Date / Time Period	Short Description
		utilization, model size and detection performance.
Chapter 2 & 3 Writing	**01 Dec 2026 – 31 Dec 2026**	Parallel writing of the literature review and methodology chapters, including EVSE cybersecurity, CICEVSE2024, preprocessing, dimensionality reduction and proposed deep-learning architectures.
Chapter 4 – Results & Discussion	**18 Dec 2026 – 31 Jan 2027**	Documentation and analysis of experimental results, comparison of all five models, dimensionality-reduction analysis and Jetson Nano deployment results.
Chapter 1 – Introduction	**25 Jan 2027 – 30 Jan 2027**	Writing of introduction, motivation, problem statement, objectives, research questions, scope and thesis organization.
Chapter 5 – Conclusion & Future Work	**31 Jan 2027 – 02 Feb 2027**	Writing of conclusions, research contributions, limitations and recommendations for future research.
Milestone 6 – Complete Draft	**03 Feb 2027**	Preliminary complete version of the thesis prepared and submitted/discussed with the supervisor.
Revision	**04 Feb 2027 – 07 Feb 2027**	Incorporation of supervisor feedback, linguistic correction, technical revisions and improvement of figures, tables and explanations.
Formal Correction	**08 Feb 2027 – 11 Feb 2027**	Final formatting, bibliography verification, figure/table numbering, references, page numbering and compliance with university requirements.
Final Validation	**12 Feb 2027 – 13 Feb 2027**	Final verification of experimental results, source-code consistency, Jetson Nano results, conclusions and thesis formatting.
Final Submission	**14 Feb 2027 – 15 Feb 2027**	Final submission of the completed scientific work to the examination office.

Overall Timeline

15 September 2026 → 15 February 2027

Approximately **22 weeks**.

Major research sequence

15 Sep – 09 Oct → Planning + Literature Review

10 Oct – 23 Oct → CICEVSE2024 Dataset Preparation

24 Oct – 05 Dec → Five Deep-Learning Pipelines

07 Dec – 31 Dec → Evaluation + Optimization + Best Model Selection

01 Jan – 24 Jan → Jetson Nano Deployment + Validation

01 Dec – 02 Feb → Parallel Thesis Writing

03 Feb – 13 Feb → Supervisor Revision + Formal Correction

14–15 Feb → Final Submission